

Grassmann Algebra

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Abstract

The algebra of space.

Euclidean geometry was put on an algebraic footing when René Descartes demonstrated how to replace ruler and compass constructions in Euclidean geometry by associating points in space with numerical coordinates, but Descartes only considered two and three dimensions. Grassmann extended this to Euclidean space of any number of dimensions with the audaciously simple rule: “The product of two points is zero if and only if they coincide.” Grassmann predated Einstein in treating all points in space of any dimension on equal footing. Grassmann’s audaciously simple notion also predated the modern definition of a vector space. Any point in space can be chosen as an origin and a vector is just the difference of a point with the origin.

Alas, Grassmann was a lowly secondary-school teacher who wrote in an abstruse style that top mathematicians of the time failed to pay attention to. It wasn’t until (Peano 1888) wrote an exposition of the theory that his ideas began to gain traction. Peano, like Descartes, only developed the theory in two and three dimensions but used a notation more accessible to mathematicians of the time in his attempt to popularize it.

Exterior Product

Grassmann considered the set E of points in Euclidean space of any (finite) dimension. The *Grassmann algebra* of E , $\mathcal{G}(E)$, is the algebra generated by the points in E over the real numbers $\mathbf{R}$ with the axiom $PQ = 0$ if and only if $P = Q$. More precisely, the Grassmann algebra is the quotient algebra $\mathcal{G}(E) = \text{Alg}(E) / \text{Alg}(\{PP \mid P \in E\})$.

The (associative) algebra over the real numbers generated by any set S , $\text{Alg}(S)$, is defined by

- $\mathbf{R} \subseteq \text{Alg}(S)$
- $S \subseteq \text{Alg}(S)$
- If $a \in \mathbf{R}$ and $x \in \text{Alg}(S)$ then $ax = xa \in \text{Alg}(S)$
- If $x, y \in \text{Alg}(S)$ then $x + y = y + x$ and xy are in $\text{Alg}(S)$
- If $x, y, z \in \text{Alg}(S)$ then $x + (y + z) = (x + y) + z$ and $x(yz) = (xy)z$.
- If $x, y, z \in \text{Alg}(S)$ then $x(y + z) = xy + xz$.
- If $x, y, z \in \text{Alg}(S)$ then $(x + y)z = xz + yz$.
- If $x \in \text{Alg}(S)$ then $0x = x0 = 0$ and $1x = x1 = x$.

Note that we do not require $xy = yx$. The associativity of addition and multiplication allows writing $x + y + z$ and xyz unambiguously.

Exercise. Show if $x \in \text{Alg}(\{PP \mid P \in E\})$ and $y \in \text{Alg}(S)$ then xy and yx are in $\text{Alg}(\{PP \mid P \in E\})$.

This shows $\text{Alg}(\{PP \mid P \in E\})$ is a two-sided ideal of $\text{Alg}(S)$ so the quotient space $\mathcal{G}(E)$ is well defined.

The product in $\mathcal{G}(E)$ is Grassmann's *exterior product*. Italian differential geometers decided to call it the *wedge product*, written as $P \wedge Q$. This was an unfortunate historical accident. The exterior product is analogous to a union of sets so $P \vee Q$ would be a more appropriate notation. Grassmann's regressive product, defined below, is analogous to the intersection of sets so $P \wedge Q$ would be the natural notation. We use a period for that, $P.Q$.

Examples

Before introducing the general theory let's consider some simple examples.

Exercise. Prove $PQ = -QP$ for $P, Q \in E$.

Hint: Use $(P + Q)(P + Q) = 0$.

The product PQ represents the oriented line determined by P and Q when $P \neq Q$. We can interpret $aP + bQ$ as the point having weight $a + b$ that is the barycentre of aP and bQ .

Let $R(t) = P + t(Q - P) = (1 - t)P + tQ$, $t \in \mathbf{R}$. Note $R(t)$ has weight 1, $R(0) = P$, and $R(1) = Q$.

Exercise Show $PQR(t) = 0$ for $t \in \mathbf{R}$.

Hint: $PQR(t) = PQ((1 - t)P + tQ)$.

This is an example of incidence involve three points. It indicates $R(t)$ meets the line determined by P and Q for all $t \in \mathbf{R}$ if $P \neq Q$.

Exercise Show $Q - P \neq R(t)$ for any $t \in \mathbf{R}$.

Since $R(t)/t = (1/t - 1)P + Q$ and $1/t - 1$ tends to -1 as t goes to infinity, we can think of $Q - P$ as a point at infinity with 0 weight. It represents the *vector* from P to Q . We can interpret $R(t) = P + t(Q - P)$ as the point from P along the vector $Q - P$ with length t .

Note $PQ(Q - P) = 0$ so the vector/point at infinity $Q - P$ is “on” the line PQ .

If $PQ = tST$ where $t \in \mathbf{R}$ and $P, Q, S, T \in E$ we say PQ and ST are *congruent* and write $\frac{PQ}{ST} = t$.

Exercise. Show $\frac{(Q-P)Q}{PQ} = -1$ and $\frac{P(Q-P)}{PQ} = 1$.

Exercise. For any R with $PQR = 0$ and $PQ \neq 0$ we have $R = \frac{PR}{PQ}P + \frac{RQ}{PQ}Q$.

Hint: If $R = aP + bQ$ then $PR = bPQ$ and $QR = aQP$.

This is an example of coordinate-free calculation using the Grassmann algebra.

David Hilbert noticed Euclid’s axioms were insufficient to determine when a point lies between two points. See (Rouse Ball 1892) for an entertaining, but incorrect, proof that all triangles are isosceles. Using Grassmann algebra we can define R is *between* P and Q if and only if $PQR = 0$, $\frac{PR}{PQ} > 0$, and $\frac{RQ}{PQ} < 1$.

A classical result from Euclidean geometry is the three medians of a triangle meet at the same point. A *median* is the line from a vertex of a triangle to the midpoint of the opposite side.

Three points $P, Q, R \in E$ determine a triangle. The midpoint of PQ is $(P + Q)/2$ and $R(P + Q)/2$ is the *median* from R to the midpoint of P and Q . The *centroid* $(P + Q + R)/3$ lies on the median.

Exercise. Show $(P + Q)R(P + Q + R) = 0$.

Replacing P, Q, R with Q, R, P and R, P, Q shows all medians of a triangle meet at the centroid.

Order Two

If there exist $P, Q \in E$ with $PQ \neq 0$ we consider the problem of when $PQ = ST$ for $S, T \in E$. Since $PQS = 0$ and $PQT = 0$ we know both S and T are on the line determined by PQ so $S = R(s)$ and $T = R(t)$ for some $s, t \in \mathbf{R}$. Since $ST = R(s)R(t) = (t - s)PQ$ we have $t - s = 1$.

Exercise. Show $PQ = TS$ if and only if $Q - P = T - S$ and either $PQT = 0$ or $PQT = 0$.

Hint: If $PQT = 0$ and $Q - P = T - S$ then $PQS = PQ(T + P - Q) = 0$.

Exercise. Given $P, Q \in E$ then $\frac{TS}{PQ}$ is the length of the vector $T - S$.

Hint: If $S = R(s)$ and $T = R(t)$ the length of $T - S$ is $t - s$.

Exercise. If E is 1-dimensional then $PQ = ST$ if and only if $Q - P = T - S$.

Hint: For any $S, T \in E$ we have $PQS = 0$ and $PQT = 0$.

Higher Dimensions

Grassmann's algebra works in any number of dimensions. Any non-zero product of points is an *extensor* (also called a *blade*, or *flag*). The number of points is the *order* of the extensor.

Congruence

If $P_0 \cdots P_k = tQ_0 \cdots Q_k$ for some non-zero $t \in \mathbf{R}$ then we write $t = \frac{P_0 \cdots P_k}{Q_0 \cdots Q_k}$ and say the extensors are *congruent*.

Independence

The points $P_0, \dots, P_k$ are *independent* if $P_0 \cdots P_k \neq 0$.

Lemma. If $P_0, \dots, P_k$ are independent and $\sum_j a_j P_j = 0$ for some $a_j \in \mathbf{R}$ then $a_i = 0$ for $0 \leq i \leq k$.

Proof. Multiply both sides by the product of $\prod_{j \neq i} P_j$. This results in $\pm a_i P_0 \cdots P_k = 0$ so $a_i = 0$ for all $0 \leq i \leq k$.

If $P_0 \cdots P_k = 0$ we say $P_0, \dots, P_k$ are *dependent*.

Lemma. If $P_0, \dots, P_k$ are dependent then there exists i and $a_j \in \mathbf{R}$, $j \neq i$, with $P_i = \sum_{j \neq i} a_j P_j$.

Exercise. Show if $P_0 \cdots P_n R = 0$ and $P_0 \cdots P_n \neq 0$ then $P = \sum_0^n \frac{P_0 \cdots P_{i-1} P P_{i+1} \cdots P_n}{P_0 \cdots P_n} P_i$.

Exercise. Show the coefficients sum to 1.

This is essentially Cramer's rule. TODO: ref

The smallest convex set containing $P_0, \dots, P_n$ is $\{\sum_i t_i P_i \mid t_i \geq 0, \sum_i t_i = 1\}$. The above exercise show P belongs to the convex hull of the points if and only if all coefficients in the sum are non-negative.

Regressive Product

We say an extensor $\Omega \in \mathcal{G}(E)$ is a *volume element* if $\Omega \neq 0$ and $P\Omega = 0$ for all $P \in E$. The *order* of E is the number of points in the extensor and the *dimension* of E is one less than its order.

The *complement* of an extensor A with respect to a volume element Ω is the unique extensor $\overline{A}$ for which $A\overline{A} = \Omega$.

TODO: Show $\overline{\overline{A}}$ is unique.

If E has dimension n and $P_0 \cdots P_n \neq 0$ then for any subset $I \subseteq \{0, \dots, n\}$ the complement of $P_I = \Pi_{i \in I} P_i$ is $\overline{P_I} = \pm \Pi_{i \notin I} P_i$ where the sign is $\frac{P_I \overline{P_I}}{P_0 \cdots P_n}$.

Exercise. Show $\overline{\overline{A}} = A$.

This can be extended linearly to $\mathcal{G}(E)$. The regressive product $A.B$ is defined by $\overline{A.B} = \overline{A} \overline{B}$

Peano, Giuseppe. 1888. *Calcolo Geometrico Secondo l'ausdehnungslehre Di h. Grassmann Preceduto Dalle Operazioni Della Logica Deduttiva*. Torino: Fratelli Bocca.

Rouse Ball, W. W. 1892. *Mathematical Recreations and Problems*. 2nd ed. London: Macmillan.